

DILEEP GUGLAVATH

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SUMMARY

Software Development Engineer (SDE-1) at Zapcom Solutions, building backend and full-stack features for a production business-travel platform. Experienced across the full software development lifecycle - API design, database schema design, backend development, testing, and frontend integration - with a focus on clean, well-documented, production-ready code. Builds independent full-stack AI products outside of work, including a retrieval-augmented generation (RAG) chat assistant and an AI-powered interview-preparation platform, using Node.js, TypeScript, React, PostgreSQL, Python, and AWS cloud infrastructure.

EXPERIENCE

Zapcom Solutions - Software Development Engineer I (SDE-1)

Jan 2026 - Present

Business Travel Platform (Client Project) | Node.js, TypeScript, Express, PostgreSQL, Prisma, Next.js

Hyderabad, India

Joined as an Intern on Jan 5, 2026; converted to a full-time role on Jul 5, 2026.

- Designed and shipped a URL shortener feature end-to-end across the backend and Next.js frontend, implementing a hybrid Base62 / SHA-256 / Snowflake-ID short-code strategy for idempotent, collision-free link generation
- Built a content-management system (CMS) from scratch - page CRUD on a builder-ready schema, a full publish/schedule/unpublish/archive lifecycle with a background scheduler for due and expired pages, and per-page, audience-based access control (RBAC)
- Debugged and resolved production issues including a CORS misconfiguration and an intermittent PostgreSQL connection-pool failure, bringing the backend test suite from failing to fully passing (405/405)
- Resolved a Prisma driver-adapter incompatibility by replacing an unsupported **\$transaction** call with a sequential-operations-and-rollback pattern preserving the same data-integrity guarantees
- Built the notification and email system end-to-end from a 19-item Azure DevOps backlog - Prisma schema and migrations, layered service integration, and AWS SES/SQS-based delivery with retry and dead-letter-queue (DLQ) handling
- Collaborate in an Agile/Scrum team using Azure DevOps and AWS CodeCommit for sprint planning, code review, and version control

PROJECTS

CBIT Guru - AI-Powered RAG Chat Assistant (B.E. Major Project)

2026

Python, FastAPI, React, Qdrant, Gemini, Cohere, Redis, Docker, GitHub Actions

- Built a retrieval-augmented generation (RAG) chatbot answering questions about a college from its own website, PDFs, and notices, with source citations, campus images, voice input/output, and replies in whatever language and script it's asked in (English, Hindi, Telugu, Hinglish, Tinglish)
- Architected the RAG pipeline - FastAPI backend, Gemini embeddings, a Qdrant vector database (cosine similarity, HNSW indexing), and Cohere command-r for grounded generation - with a two-tier response cache (exact match, then cosine paraphrase match) to cut redundant LLM API calls
- Hardened for production with concurrency caps, per-call timeouts, and a circuit breaker so a provider outage fails fast instead of cascading, plus Prometheus metrics and request-ID-correlated logging
- Built a 43-assertion offline smoke-test suite against an in-memory vector database with stubbed providers, running on every push via GitHub Actions alongside linting and a frontend build, at zero API cost

InterviewEdge - AI-Powered Interview Preparation Platform (MERN Stack)

2026

React 19, Vite, Tailwind CSS, Node.js, Express 5, MongoDB, Redis, JWT | interviewedge-ten.vercel.app

- Built a full-stack app with JWT + bcrypt authentication and ownership-based authorization on every route; features include resume-aware question generation, timed mock interviews with AI scoring, and a per-role leaderboard with a privacy opt-out
- Integrated Google Gemini 2.5 Flash as the primary LLM with a Groq Llama 3.3 70B fallback and a 4-strategy JSON-repair layer for reliable structured AI output
- Designed a three-tier cache (CDN, then Redis, then MongoDB) with normalized keys and tiered TTLs to cut redundant AI calls and response latency
- Led a full production-readiness audit - hardened CORS, centralized error handling, fixed authorization (IDOR) bugs, and rate-limited auth and AI endpoints
- Wrote 90 Jest + Supertest tests (unit, integration, load, and resilience), with full technical docs, a Postman collection, and an n8n automation workflow

Additional Projects

- **Stock Price Prediction (AAPL)** - Linear regression and LSTM models with a full ETL pipeline; 85% directional accuracy, 15% lower RMSE than baseline
- **Algorithmic Trading Strategy (Pine Script)** - ATR-based trailing-stop/crossover strategy on TradingView, backtested on 5+ years of NIFTY index data with automated buy/sell alerts

SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, SQL

Backend: Node.js, Express.js, REST API Design, Prisma ORM, JWT Auth, Zod

AI & Machine Learning: RAG Pipelines, LLM Integration (Gemini, Cohere, Groq), Vector Databases (Qdrant), Prompt Engineering, TensorFlow, scikit-learn, pandas, OpenCV

Databases & Caching: PostgreSQL, MongoDB, Redis, Qdrant

Frontend: Next.js, React, Tailwind CSS, Vite

Cloud & DevOps: AWS (EC2, RDS, S3, Lambda, ECS, IAM, VPC, SES, SQS, SNS), Docker, Git, GitHub Actions (CI/CD), Azure DevOps

Testing & Tooling: Jest, Supertest, Unit Testing, Integration Testing, Postman, Swagger/OpenAPI, VS Code

Practices: Agile/Scrum, System Design, TDD, Layered Architecture, Technical Documentation

EDUCATION

Chaitanya Bharathi Institute of Technology, Hyderabad	2023 - 2026
B.E. in Artificial Intelligence and Data Science - CGPA: 8.05/10	
Mahaveer Institute of Science and Technology, Hyderabad	2020 - 2023
Diploma in Computer Science Engineering (SBTET) - CGPA: 8.88/10	

CERTIFICATIONS

- Oracle Cloud Infrastructure - AI Foundations Associate (Apr 2024)
- Data Science Foundation Certification - Python, EDA, Statistics, Data Visualization
- Self-Directed AWS Learning Path - EC2, RDS, S3, Lambda, ECS, IAM, VPC (2026)